Fast AI/ML Learning Path (8–12 Weeks)

Phase 1: Mathematics Essentials (1 Week)

Focus only on the math required for ML.

Learn:

- Linear Algebra
  - Vectors
  - Matrices
  - Matrix multiplication
  - Dot product
- Statistics & Probability
  - Mean, Median, Variance
  - Standard Deviation
  - Probability distributions
  - Bayes' Theorem
- Calculus Basics
  - Derivatives
  - Gradient Descent

Important formulas:

Gradient Descent
$\theta = \theta - \alpha \nabla J(\theta)$

Bayes Theorem

$P(A|B)=\frac{P(B|A)P(A)}{P(B)}$

Phase 2: Python for Data Science (1 Week)

Learn:

- NumPy
- Pandas
- Matplotlib
- Seaborn (commonly used, though Matplotlib is the foundation)

Practice:

- Load datasets
- Clean data

- Visualize trends
- Feature engineering basics

Projects:

- Titanic Dataset Analysis
- Netflix Data Analysis
- Sales Prediction Dataset

Phase 3: Core Machine Learning (2–3 Weeks)

Supervised Learning

Learn:

- Linear Regression
- Logistic Regression
- Decision Trees
- Random Forest
- Support Vector Machines (SVM)
- KNN

Unsupervised Learning

Learn:

- K-Means Clustering
- Hierarchical Clustering
- PCA

Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC

Important formula:

Linear Regression

$$y = mx + b$$

Tools:

- Scikit-learn
- Jupyter Notebook

Projects:

1. House Price Prediction
2. Student Performance Prediction
3. Customer Segmentation
4. Loan Approval Prediction

Phase 4: Deep Learning (2 Weeks)

Learn:

- Artificial Neural Networks (ANN)
- Activation Functions
    - ReLU
    - Sigmoid
    - Softmax
- Backpropagation
- Optimization

Framework:

- PyTorch (recommended)
    - Industry and research standard
- TensorFlow/Keras (optional)

Projects:

- Handwritten Digit Recognition (MNIST)
- Fashion Item Classification

Phase 5: Computer Vision (1–2 Weeks)

Learn:

- Image Processing Basics
- Convolutional Neural Networks (CNNs)
- Transfer Learning

Libraries:

- OpenCV
- PyTorch

Projects:

- Face Mask Detection
- Cat vs Dog Classifier
- Face Shape Detection (matches your glasses recommendation project)

Phase 6: NLP & LLMs (2 Weeks)

Learn:

- Text Preprocessing
- Tokenization
- Word Embeddings
- Transformers
- Attention Mechanism
- Large Language Models

Libraries:

- Hugging Face Transformers
- LangChain
- OpenAI APIs

Projects:

- Sentiment Analysis
- Resume Analyzer
- Chatbot
- Document Q&A System

Phase 7: MLOps & Deployment (1 Week)

Since you're a full-stack developer, this is where you'll gain a strong advantage.

Learn:

- FastAPI

- Docker
- Model Deployment
- ML Pipelines
- Vector Databases
- RAG (Retrieval-Augmented Generation)

Projects:

- Deploy ML model with FastAPI
- Deploy AI app on Vercel + Railway/AWS
- Build an AI SaaS application

Portfolio Projects (Highly Recommended)

Beginner

- House Price Predictor
- Spam Email Classifier
- Movie Recommendation System

Intermediate

- Customer Segmentation
- Face Shape Detection
- Resume Screening System

Advanced

- AI Interview Assistant
- Multilingual AI Chat System
- RAG-based Document Chatbot
- AI Agent using LangGraph
- Medical Diagnosis Assistant

Best Resources

Machine Learning

- Scikit-learn Documentation
- Kaggle Learn

Deep Learning

- PyTorch Documentation
- DeepLearning.AI

NLP & LLMs

- Hugging Face Course

Mathematics

- 3Blue1Brown Linear Algebra Series

Fastest Route to Become Job-Ready

Given your background, the shortest high-impact route is:

Python → NumPy/Pandas → Scikit-learn → PyTorch → Transformers → RAG → FastAPI → AI Projects

If you dedicate **2–3 hours daily**, you can build a strong AI/ML portfolio and become interview-ready for junior AI/ML or AI Engineer roles in about **3 months**.